Generic, Excel-based dynamic simulation tool | Developed by: M.L. Jon

last modified 9/19/99

This workbook contains templates for worksheets and Visual Basic code to run dynamic simulation models and save the outputs for viewing and possible analyses

Initial Conditions

Place initial conditions and process parameters in this worksheet, keeping track of cells where the values are located

Example: a Swomee-swan population model

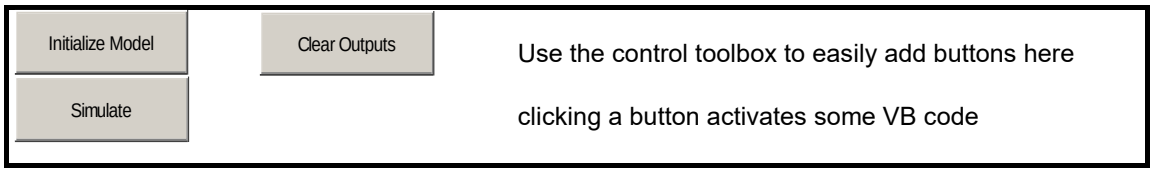
Initial number of swans	4	**this initial condition is in cell(11,4)**
Birth rate parameter 1	2	**these four parameters are in cells(13,4) through (16,4)**
Birth rate parameter 2	0.0046	
Death rate parameter 1	1	
Death rate parameter 2	50	



Run Options

This sheet contains the buttons you press to run the model and any run opti
specify. Always specify the number of time steps….

If you also specify the range of cells for your outputs, the clear outputs butt
cells you want to have erased. By default 200 rows by 8 columns will be e
time step column.



Number of time steps to simulate: 50
Range of cells for outputs: A6:D205

ons you may want to

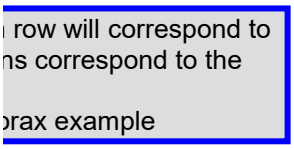
on will only erase the
rased, including the

Outputs

This is where the model outputs (indicator variables you select) will go. Each
the state of the indicator(s) at the end of a particular time step. The colum
indicators you specify in your model code.
You can also inlcude calculated fields here - see column I in the Lo

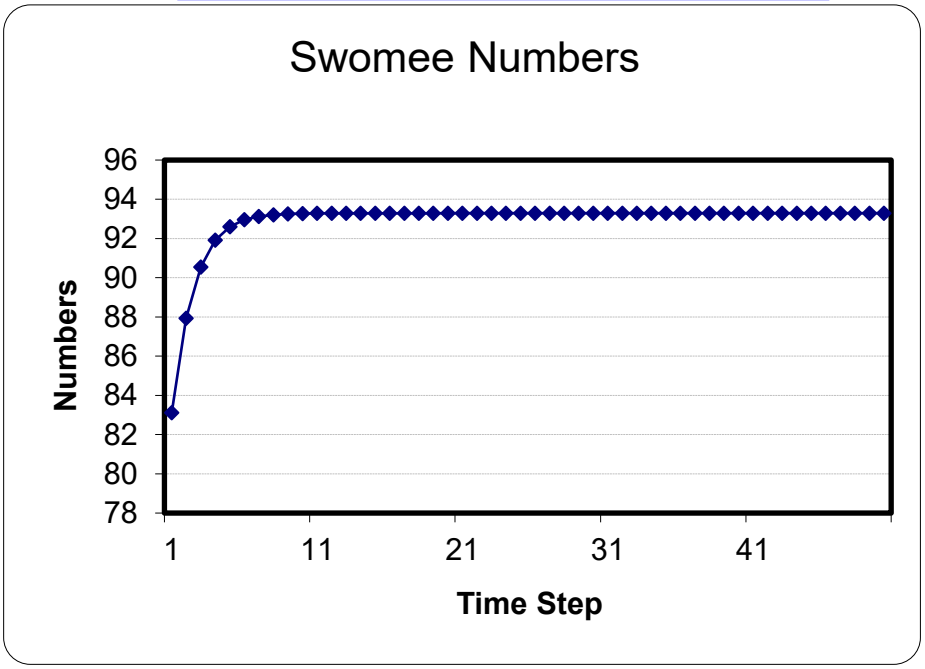
Time Step	Swomees	BirthRate	DeathRate
1	83.11653	1.416441	0.6
2	87.92702	1.364532	0.624389
3	90.55119	1.334668	0.637489
4	91.91574	1.318654	0.644258
5	92.60725	1.310403	0.647678
6	92.95305	1.306241	0.649387
7	93.12481	1.304165	0.650235
8	93.20983	1.303135	0.650655
9	93.25185	1.302626	0.650862
10	93.27261	1.302374	0.650964
11	93.28285	1.30225	0.651015
12	93.2879	1.302188	0.65104
13	93.29041	1.302158	0.651052
14	93.29164	1.302143	0.651058
15	93.29224	1.302135	0.651061
16	93.29254	1.302132	0.651063
17	93.29269	1.30213	0.651064
18	93.29276	1.302129	0.651064
19	93.29279	1.302129	0.651064
20	93.29282	1.302129	0.651064
21	93.29282	1.302128	0.651064
22	93.29282	1.302128	0.651064
23	93.29282	1.302128	0.651064
24	93.29282	1.302128	0.651064
25	93.29282	1.302128	0.651064
26	93.29282	1.302128	0.651064
27	93.29282	1.302128	0.651064
28	93.29282	1.302128	0.651064
29	93.29282	1.302128	0.651064
30	93.29282	1.302128	0.651064
31	93.29282	1.302128	0.651064
32	93.29282	1.302128	0.651064
33	93.29282	1.302128	0.651064
34	93.29282	1.302128	0.651064
35	93.29282	1.302128	0.651064
36	93.29282	1.302128	0.651064
37	93.29282	1.302128	0.651064
38	93.29282	1.302128	0.651064
39	93.29282	1.302128	0.651064
40	93.29282	1.302128	0.651064
41	93.29282	1.302128	0.651064
42	93.29282	1.302128	0.651064
43	93.29282	1.302128	0.651064
44	93.29282	1.302128	0.651064
45	93.29282	1.302128	0.651064
46	93.29282	1.302128	0.651064
47	93.29282	1.302128	0.651064
48	93.29282	1.302128	0.651064
49	93.29282	1.302128	0.651064

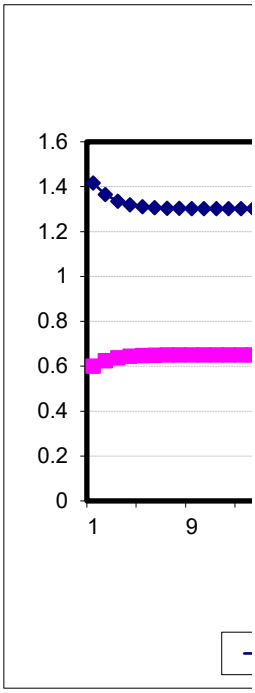
50	93.29282	1.302128	0.651064

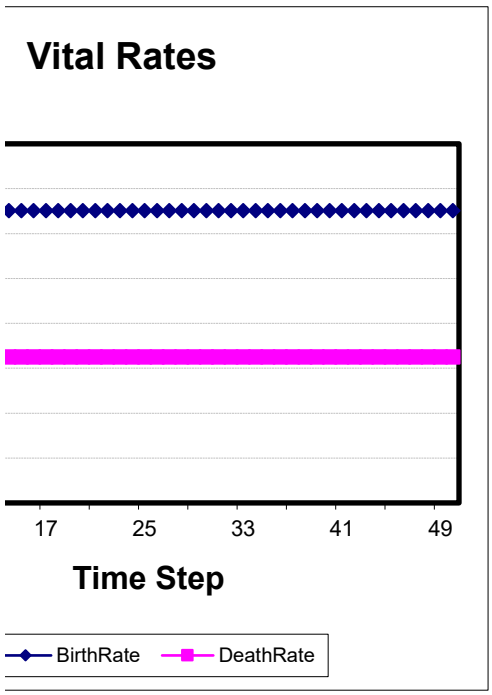

row will correspond to
ns correspond to the
rax example

Charts

Add any charts you'd like to use to view model results to this worksheet. Generally, you will be using line charts with column 1 of the Outputs worksheet as the x-axis series.






Vital Rates
17
25
33
41
49
Time Step
BirthRate
DeathRate